Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 11

Roll No :495

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Progrmas based on Python Data Structure + UDF + File Handling

Practical 1: Count Lines, Words and Characters

Problem Definition:
Write a Python program to read text from a file and count the total number of lines, words, and characters. The result should be written to an output file.

In [8]:
```
with open("input.txt","r") as fp:
    txt = fp.read()

lines = txt.split("\n")
lines_count = len(lines)

words = txt.split()
words_count = len(words)

char = len(txt)

with open("output.txt","w") as fp1:
    fp1.write(f"No of Lines:{lines_count}\n")
    fp1.write(f"No of Words:{words_count}\n")
    fp1.write(f"No of Char:{char}\n")
```

Practical 2: Search a Word in a File

Problem Definition:
Write a Python program to search a given word in a text file and count its occurrences. Store the result in an output file.

In [10]:
```
search_word = input("Enter Word to Search:")

with open('input.txt','r') as fp:
    txt = fp.read()

txt = txt.lower()
search_word = search_word.lower()

word_count = txt.split().count(search_word)

with open('output.txt','w') as fp1:

    fp1.write(f"Words Searched:{search_word}\n")
    fp1.write(f"Occurence:{word_count}\n")
```

Practical 3: Student Marks and Average

Problem Definition:
Write a Python program to read roll number, name, and marks of students from a file. Store marks in a list, calculate the average using a user-defined function, and write student details along with average marks to an output file.

In [7]:
```
def calculate_average(marks_list):
    return sum(marks_list) / len(marks_list)

with open("input.txt", "r") as file:
    students = file.readlines()

with open("output.txt", "w") as file:
    for student in students:
        roll, name, marks = student.split()
        marks = int(marks)

        avg = calculate_average([marks])

        file.write(f"Roll No: {roll}, Name: {name}, Marks: {marks}, Average: {avg}\

print("Student averages calculated.")
```

```
Student averages calculated.
```

Practical 4: Maximum and Minimum using Tuple

Problem Definition:
Write a Python program to read numbers from a file, store them in a tuple, and find the maximum and minimum values. Write the result to an output file.

In [34]:
```
with open("input.txt", "r") as file:
    data = file.read()

numbers = tuple(map(int, data.split()))

maximum = max(numbers)
minimum = min(numbers)

with open("qutput.txt", "w") as file:
    file.write(f"Numbers: {numbers}\n")
    file.write(f"Maximum value: {maximum}\n")
    file.write(f"Minimum value: {minimum}\n")

print("Maximum and minimum values written ")
```

```
Maximum and minimum values written
```

Practical 5: Remove Duplicate Words using Set

Problem Definition:
Write a Python program to read words from a file and remove duplicate words using a set. Store the unique words in an output file.

In [14]:
```
with open("que5.txt", "r") as file:
    text = file.read()

words = text.split()

unique_words = set(words)

with open("que5_op.txt", "w") as file:
    for word in unique_words:
        file.write(word + " ")
```

Practical 6: Word Frequency using Dictionary

Problem Definition:
Write a Python program to count the frequency of each word in a text file using a dictionary and write the result to an output file.

In [ ]:
```
with open("input6.txt", "r") as file:
    content = file.read()

content = content.lower()

words = content.split()
```

```
word_freq = {}

for word in words:
    if word in word_freq:
        word_freq[word] += 1
    else:
        word_freq[word] = 1

with open("output6.txt", "w") as file:
    file.write("Word Frequencies:\n")
    for word, count in word_freq.items():
        file.write(f"{word} : {count}\n")

print("Word frequency written to output6.txt successfully!")
```

Practical 7: Student Result and Grade Calculation

Problem Definition:
Write a Python program to read roll number, name, and marks of students from a file. Use a user-defined function to determine grade (A/B/C/Fail) and write the student result to an output file.

Grade Criteria:

- Marks ≥ 75 : Grade A
- Marks ≥ 60 : Grade B
- Marks ≥ 40 : Grade C
- Marks < 40 : Fail

In [28]:

```
Thank You
```

In [ ]: